MD ASIF KHAN NIRJON

Undergraduate Researcher | Mechanical Engineering

Department of Mechanical Engineering, Hajee Mohammad Danesh Science & Technology University (HSTU)
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RESEARCH INTERESTS

Computational Fluid Dynamics (CFD), Multiphase Flows & Volume of Fluid (VOF) Formulations, Vehicle External Aerodynamics & Drag Reduction, Aeroacoustics, Topology Optimization, Structural & Dynamic Analysis, Finite Element Analysis (FEA), and Thermal-Fluid Systems.

EDUCATION

Hajee Mohammad Danesh Science and Technology University (HSTU)

Dinajpur, Bangladesh
Bachelor of Science (B.Sc.) in Mechanical Engineering | Expected Graduation: August 2027

- **Cumulative GPA:** 3.3 / 4.00
- **Core Coursework:** Fluid Mechanics I & II, Heat & Mass Transfer, Thermodynamics, Aerodynamics, Numerical Methods in Engineering, Mechanics of Materials, Machine Design, Automobile Engineering, Power Plant Engineering.

JOURNAL PUBLICATIONS & MANUSCRIPTS UNDER REVIEW

1. Md Imran Khan, Rita Es souissi, **Md Asif Khan Nirjon**, Rifath Bin Hossain, Di Changan, and Peng Peng. "Perforated-Support Dorsal-Washing Subsystem for Assistive Bathing of Bedridden Users: Geometric, Multiphase CFD, and Mannequin Evaluation." *Under Review, Elsevier* (2026).
2. Rita Es souissi, Md Imran Khan, Rifath Bin Hossain, **Md Asif Khan Nirjon**, and Xiaoli Wu. "An Integrated Structural and Dynamic Evaluation Framework for a Topology-Optimized Lightweight Robotic Gripper." *Under Review, Progress in Engineering Science* (2026).

HONORS & ACADEMIC DISTINCTIONS

- **Government Junior Scholarship (JSC):** Awarded competitive merit scholarship by the Ministry of Education & BISE, Bangladesh for academic distinction in the Junior School Certificate Examination (2016).
- **Government Primary Scholarship (PSC):** Awarded merit scholarship by the Directorate of Primary Education (DPE), Bangladesh based on academic excellence in the Primary School Certificate Examination (2013).

RESEARCH EXPERIENCE & ACADEMIC PROJECTS

Undergraduate Thesis & Research

2026 – Present
Supervisor: Dr. K. H. Nazmul Ahsan, Associate Professor, Department of Mechanical Engineering, HSTU

- **Topic:** *To be finalized in consultation with thesis supervisor* (Domain: Computational Fluid Dynamics, Thermal-Fluid Systems & Mechanical Design).
- Conducting comprehensive preliminary literature reviews on aerodynamic optimization, boundary-layer dynamics, and computational turbulence modeling.
- Formulating numerical validation frameworks and CAD modeling pipelines using SOLIDWORKS and ANSYS Fluent in preparation for formal thesis proposal approval.

Multiphase CFD Modeling for Assistive Patient Bathing Subsystem

2026
International Collaborative Research Project (with Nanjing University of Science & Technology) | Co-Author

- Simulated high-speed multiphase flow using the Volume of Fluid (VOF) model in ANSYS Fluent with PISO pressure-velocity coupling and HRIC interface capturing to evaluate transient water impingement on human dorsal anatomy.
- Quantified dorsal water coverage thresholds (achieving ~89.1% coverage at 0.20 s) and optimized manifold nozzle discharge constraints.

Topology Optimization & Dynamic Evaluation of Lightweight Robotic Gripper

2026
Collaborative Research Publication Project | Co-Author

- Performed 3D structural CAD modeling, mesh generation, and FEA static/dynamic evaluations for a lightweight gripper mechanism.

- Assessed modal and structural stress distributions to validate weight minimization under target gripping load scenarios.

Agricultural Sprayer Machine Mechanical CAD Design

2025 – 2026
Freelance CAD Project

- Developed complete 3D mechanical part models and comprehensive assembly of a ground-based agricultural sprayer in SOLIDWORKS.
- Modeled frame structures and nozzle mounting linkages to ensure operational mechanical clearance and component fit.
- Prepared manufacturing-ready 2D technical drawings, exploded assembly views, and bill of materials (BOM).

Autonomous High-Speed Soccer Robot Mechanical Chassis Design

October 2023
Participant & Chassis Design Lead • IGNITION 2023 (National Mechanical Festival), KUET

- Engineered an omnidirectional chassis layout with low center-of-gravity in SOLIDWORKS for high-speed robotic maneuvering in the Soccer Bot Competition segment.
- Integrated motor brackets, kicker mechanics, and shock-dampening sensor brackets for dynamic arena operations.

INDUSTRIAL EXPERIENCE & PROFESSIONAL TRAINING

Training Institute for Chemical Industries (TICI)

June 2026
Mechanical Engineering Trainee • Industrial Technology on Mechanical Engineering
Central Training Institute of Bangladesh Chemical Industries Corporation (BCIC) | Polash, Narsingdi

- Completed intensive 3-week hands-on industrial training on process plant machinery, chemical pumps, compressors, steam turbines, piping networks, and plant safety standards.
- Achieved Grade A+ in the final comprehensive technical evaluation.

Data Analysis with Python and MATLAB • EDGE Project, ICT Division

Jul 2025 – Aug 2025
Conducted by Faculty of Computer Science & Engineering (CSE), HSTU | Dinajpur, Bangladesh

- Completed 80 hours of professional coursework on numerical algorithms, matrix computing, data analysis, and plotting using Python and MATLAB under the Enhancing Digital Government and Economy (EDGE) Project, Bangladesh Computer Council (BCC).

National Skill Standard Basic: Computer Office Application • BTEB

Jan 2019 – Jun 2019
Bangladesh Technical Education Board (BTEB) • 360 Hours (6 Months) | Netrakona, Bangladesh

- Earned Grade A+ in professional office productivity software, data sheets, and documentation workflows.

TECHNICAL EXPERTISE

Domain	Core Competencies & Software
Simulation & CFD	ANSYS Fluent (multiphase VOF, k-epsilon, k-omega SST, boundary layer flows, meshing), ANSYS CFX, ANSYS Mechanical (Static Structural FEA)
CAD & Visualization	SOLIDWORKS (Advanced Part Modeling, Complex Assembly, Sheet Metal, Motion Analysis), KeyShot Photorealistic Rendering
Programming & Computing	Python (Data Analysis), MATLAB (Numerical Computing, Optimization, Plotting), Linux (Ubuntu)
Technical Tools & Formats	Microsoft Office Suite (Excel, Word, PowerPoint - BTEB Certified A+), Adobe Illustrator, Adobe Photoshop, Technical Documentation

INTERNATIONAL WORKSHOPS & CERTIFICATIONS

- **Advanced SOLIDWORKS Workflows & Parametric Modeling:** International Technical Seminar & Workshop, SolidWorks User Group Network (SWUGN) & Nairobi City SWUGN (2025).
- **Modifying Geometry Using Complex Parts & Advanced Mate Optimization:** SolidWorks User Group Network (SWUGN) Certification (2025).

EXTRACURRICULAR LEADERSHIP & AFFILIATIONS

- **Active Member:** SolidWorks User Group Bangladesh (SWUG Bangladesh)

- **Executive / Active Member:** HSTU Photography Society

REFERENCES

1. **Dr. K. H. Nazmul Ahsan**
   Associate Professor, Department of Mechanical Engineering
   Hajee Mohammad Danesh Science & Technology University (HSTU)
   (Undergraduate Thesis Supervisor)
2. **Academic / Departmental Reference**
   Department of Mechanical Engineering
   Hajee Mohammad Danesh Science & Technology University (HSTU)
   Dinajpur-5200, Bangladesh
   Contact details available upon request